

Advanced Visuals in Power BI

Practical Assignment - Sales Dataset

Dataset	Sales_Dataset.xlsx
Rows	250
Date Range	2021-01-14 to 2024-12-29
Total Sales	Rs. 2,302,018
2023 Sales	Rs. 726,999
India Sales	Rs. 2,300,843

This report presents each assignment question followed by a Power BI-style solution, including filters, slicers, drill-down, drill-through, measures and visuals based on the supplied dataset.

Question 1: What is the difference between a Basic Filter, Advanced Filter, and a Slicer in Power BI?

Solution

A Basic Filter selects one or more existing values from a field. An Advanced Filter applies conditions such as greater than, contains, starts with, or a date range. A Slicer is an on-canvas interactive visual that lets report users change the filter context themselves.

Dataset result: Basic and Advanced Filters are configured in the Filters pane, while a Slicer is placed directly on the report canvas.

Question 2: Using the Sales dataset, apply a Basic Filter to display only the sales data for the year 2023.

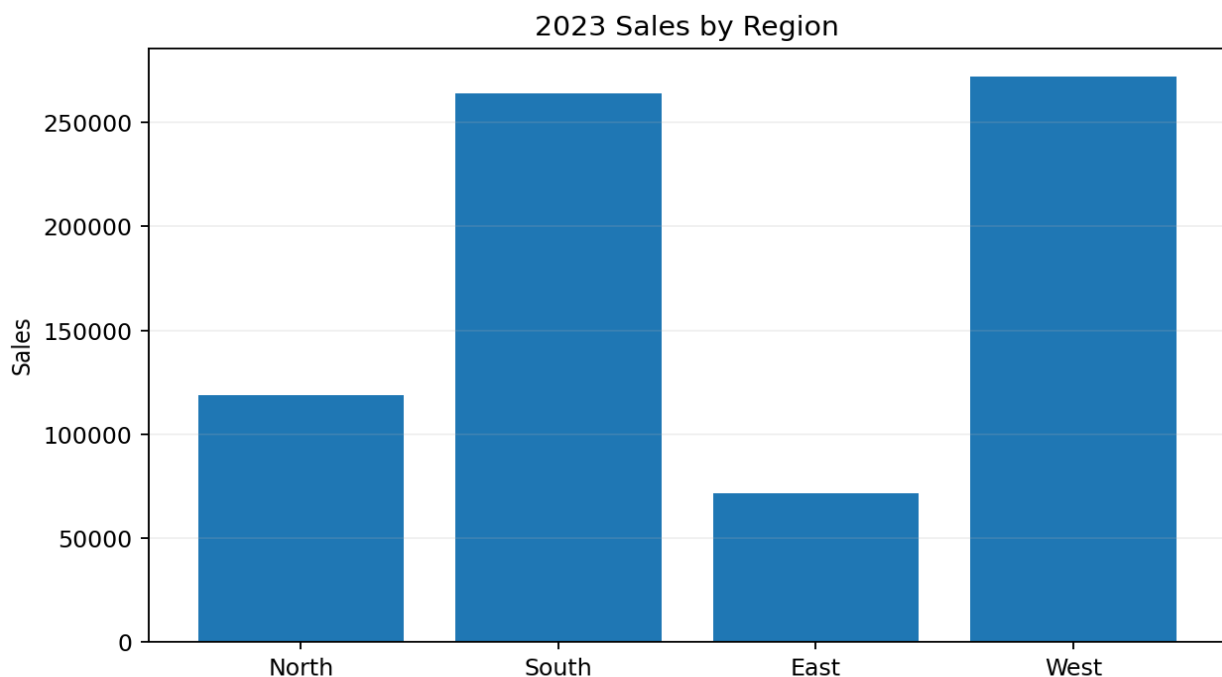
Solution

The Date field is filtered to Year = 2023 using a Basic Filter.

Power BI implementation

1. Add Date to the Filters pane.
2. Choose Basic filtering and select only 2023.
3. All visuals on the chosen filter scope update to 2023 data.

Dataset result: 2023 contains 80 orders with total sales of Rs. 726,999.



Question 3: Create a Pie Chart of Sales by Product and add Slicer that allows users to filter the dataset by Region.

Solution

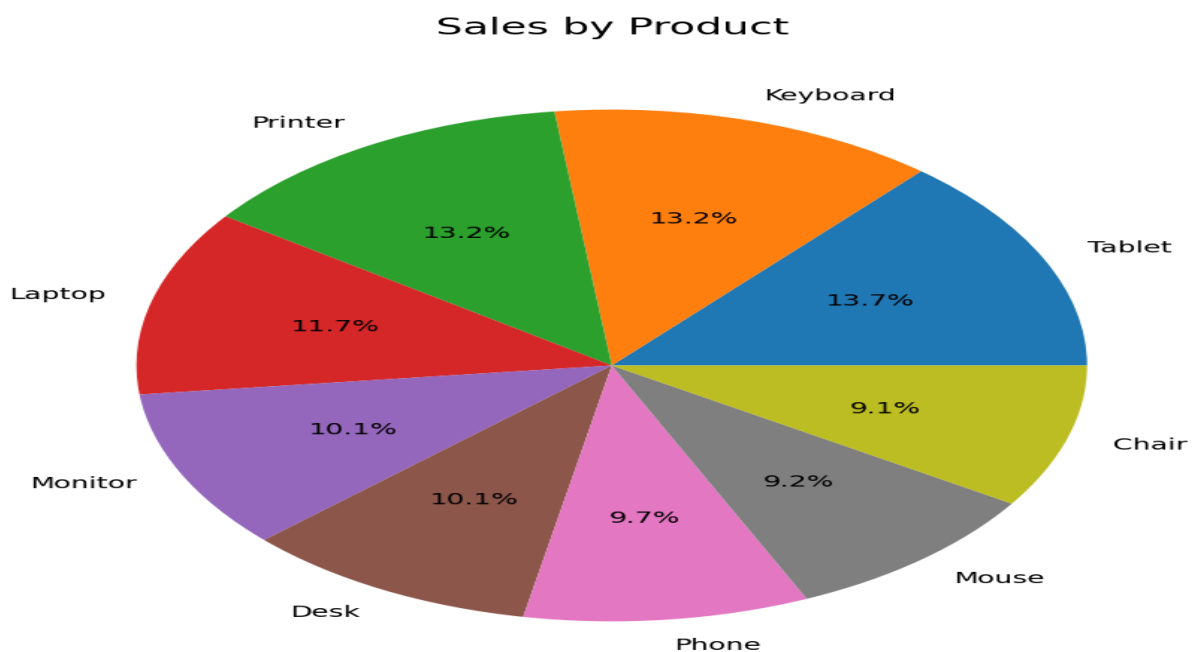
A Pie Chart is created with Product as the legend/category and Sales as Values. A Region slicer is added so users can interactively filter the chart.

Power BI implementation

1. Insert a Pie chart.
2. Drag Product to Legend and Sales to Values.
3. Insert a Slicer and drag Region into the slicer field.
4. Verify Edit interactions so the slicer filters the Pie Chart.

Dataset result: The highest total product sales are for Tablet at Rs. 315,818.

Product	Total Sales
Tablet	Rs. 315,818
Keyboard	Rs. 304,052
Printer	Rs. 303,220
Laptop	Rs. 269,552
Monitor	Rs. 233,383
Desk	Rs. 231,793
Phone	Rs. 223,560
Mouse	Rs. 211,644
Chair	Rs. 208,996



Question 4: What is a Visual-Level Filter, and how does it differ from Page-Level and Report-Level Filters?

Solution

A Visual-Level Filter affects only one visual. A Page-Level Filter affects every visual on the current report page. A Report-Level Filter affects all visuals across every page in the report.

Dataset result: Filter scope becomes broader from Visual -> Page -> Report.

Question 5: Apply a Report-Level Filter so that all visuals in the Power BI report display sales only from India.

Solution

Country is placed under Filters on all pages and only India is selected.

Power BI implementation

1. Open the Filters pane.
2. Drag Country into Filters on all pages.
3. Select India and clear other values.

Dataset result: India contributes Rs. 2,300,843 of Rs. 2,302,018 total sales. The dataset also contains 4 Dubai rows.

Question 6: What is the role of Drill-Down in Power BI? How does it improve data exploration?

Solution

Drill-Down lets users move from a summarized level of a hierarchy to a more detailed level without leaving the current visual. It improves exploration by allowing users to start with the big picture and progressively inspect the factors behind a value.

Dataset result: Typical examples are Year -> Quarter -> Month -> Day and Country -> Region -> Product Category.

Question 7: Using the dataset, create a hierarchy drill-down from Country -> Region -> Product Category to analyze sales at different levels.

Solution

A custom hierarchy is created using Country, Region, and Category. Sales is used as the value in a column chart or matrix.

Power BI implementation

1. In the Fields pane, create a hierarchy with Country at the top.
2. Add Region below Country and Category below Region.
3. Place the hierarchy on the Axis and Sales in Values.
4. Enable Drill-Down and select a Country, then Region, then Category.

Dataset result: At the Region level, West has the highest total sales at Rs. 625,797.

Region	Total Sales
West	Rs. 625,797
South	Rs. 590,265
North	Rs. 550,844
East	Rs. 535,112

Question 8: Implement Drill-Through so that selecting a Product navigates to a detailed report page showing region-wise sales.

Solution

A dedicated Product Details page is created. Product is added to the Drill-through field well, and a region-wise sales chart is placed on that page.

Power BI implementation

1. Create a new page named Product Details.
2. Add Product to the Drill-through filters section.
3. Add a bar chart with Region on Axis and Sales as Values.
4. On the source page, right-click a product data point -> Drill through -> Product Details.

Dataset result: The destination page inherits the selected Product context and displays its sales split by Region.

Question 9: Add a Slicer for Product Category and observe how it interacts with the drill-down hierarchy.

Solution

A Category slicer is added to the report page. The slicer changes the filter context of the hierarchy visual, so only the selected category remains available while drilling through Country and Region levels.

Power BI implementation

1. Insert a Slicer visual.
2. Use Category as the slicer field.
3. Select Electronics, Accessories, Office, or Furniture and test the hierarchy.

Dataset result: Category totals are: Electronics Rs. 808,930, Accessories Rs. 749,079, Furniture Rs. 440,789, Office Rs. 303,220.

Question 10: Explain the difference between Drill-Down and Drill-Through in Power BI using a real-world example.

Solution

Drill-Down explores deeper levels within the same visual and page. Drill-Through navigates to another report page while passing the selected value as a filter. Example: drill down from Year to Month in a sales chart; drill through from a selected Laptop product to a Product Details page showing its regional sales, customers and orders.

Dataset result: Drill-Down = deeper hierarchy on the same visual; Drill-Through = contextual navigation to a detail page.

Question 11: Use the Drill-Down feature to analyze sales trends from Year -> Quarter -> Month -> Day.

Solution

Power BI automatically provides a Date Hierarchy when Date is used. The hierarchy is configured as Year, Quarter, Month and Day and used on a line chart with Sales as Values.

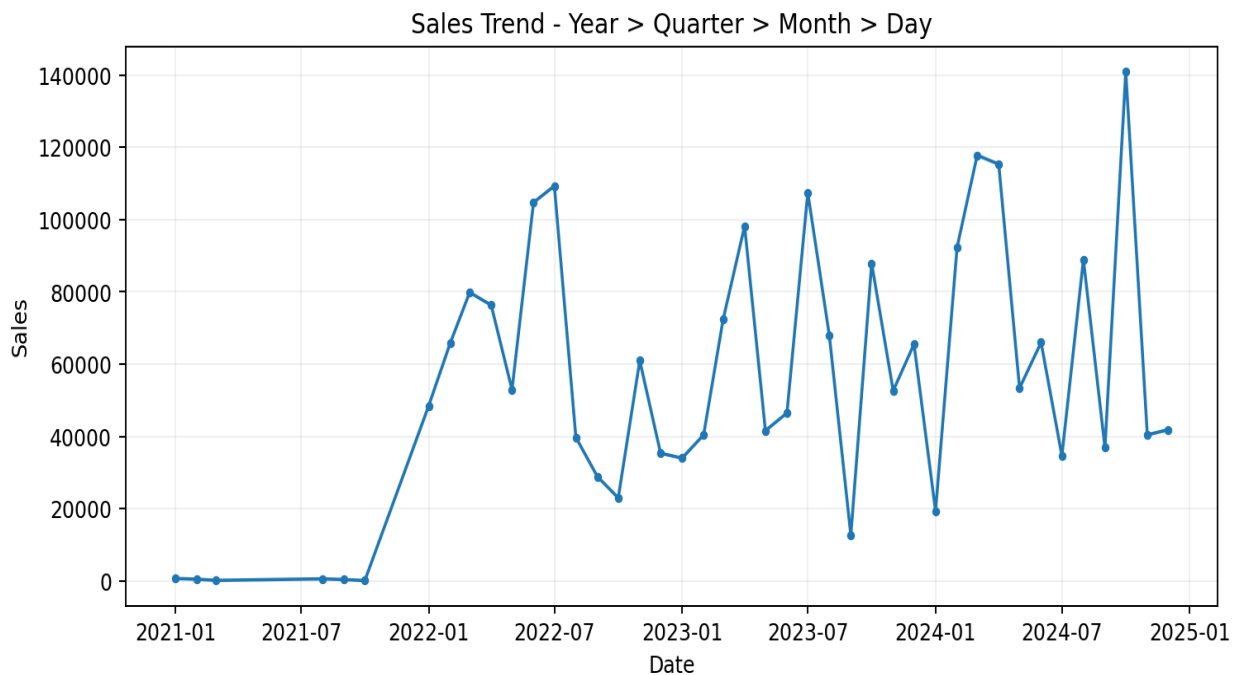
Power BI implementation

1. Insert a Line Chart.
2. Add Date Hierarchy to the X-axis.

3. Add Sales to Values.

4. Use the drill icons to move from Year to Quarter, Month and Day.

Dataset result: The dataset spans 2021-2024 and supports all four date hierarchy levels.



Question 12: Apply a Measure-Based Filter to display only those products whose total sales exceed Rs. 50,000. Hint: Create a Total Sales measure and apply the filter using the measure, not the Sales column.

Solution

A Total Sales measure is created and then used as a visual-level measure filter with the condition is greater than 50000.

Power BI implementation

1. Create the Total Sales measure.
2. Build a table or bar chart using Product and Total Sales.
3. Drag Total Sales to Filters on this visual.
4. Set Advanced filtering -> is greater than -> 50000.

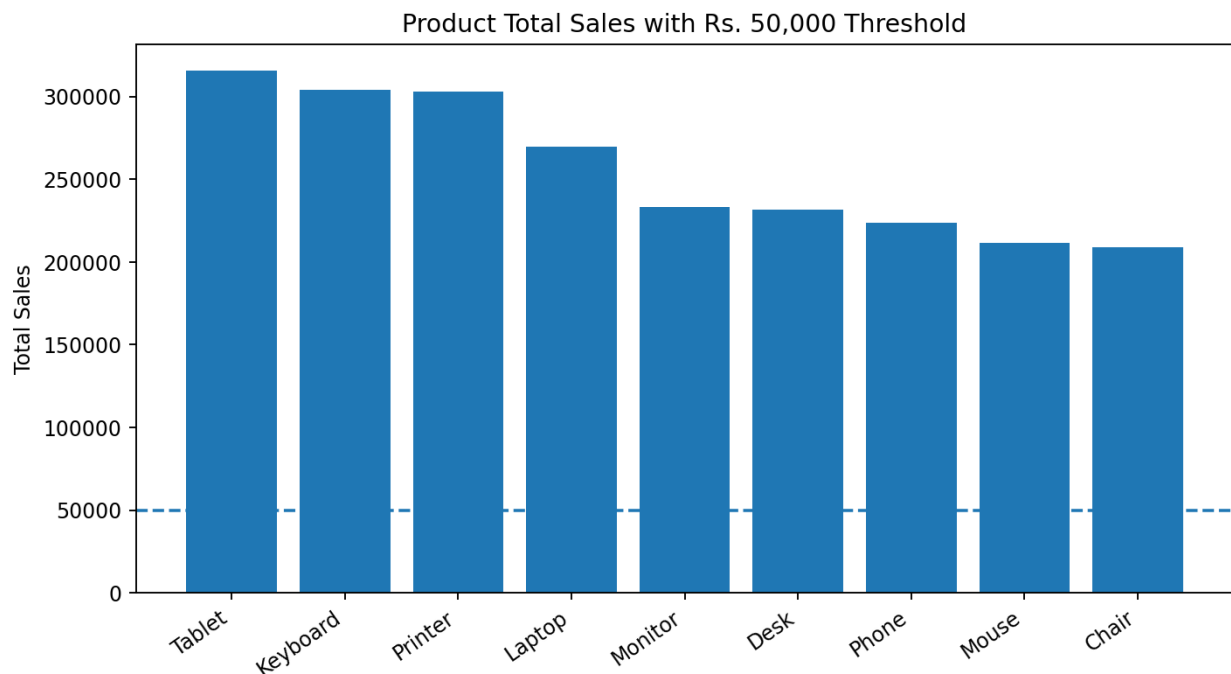
DAX

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Total Sales = SUM('Sheet1'[Sales])
```

Dataset result: 9 products satisfy the threshold in this dataset.

Product	Total Sales
Tablet	Rs. 315,818
Keyboard	Rs. 304,052
Printer	Rs. 303,220
Laptop	Rs. 269,552
Monitor	Rs. 233,383

Product	Total Sales
Desk	Rs. 231,793
Phone	Rs. 223,560
Mouse	Rs. 211,644
Chair	Rs. 208,996



Question 13: What is the "Keep All Filters" option in Drill-Through? How does it affect user interaction?

Solution

Keep All Filters passes the full current filter context from the source page to the drill-through page, not just the drill-through field. This preserves selections such as Region, Category, Date, and other active filters, providing a more contextual detail view.

Dataset result: When disabled, the destination primarily receives the configured drill-through field; when enabled, other relevant source filters are also retained.

Question 14: Create a custom Drill-Through page where selecting a Region shows a detailed view of sales trends over time.

Solution

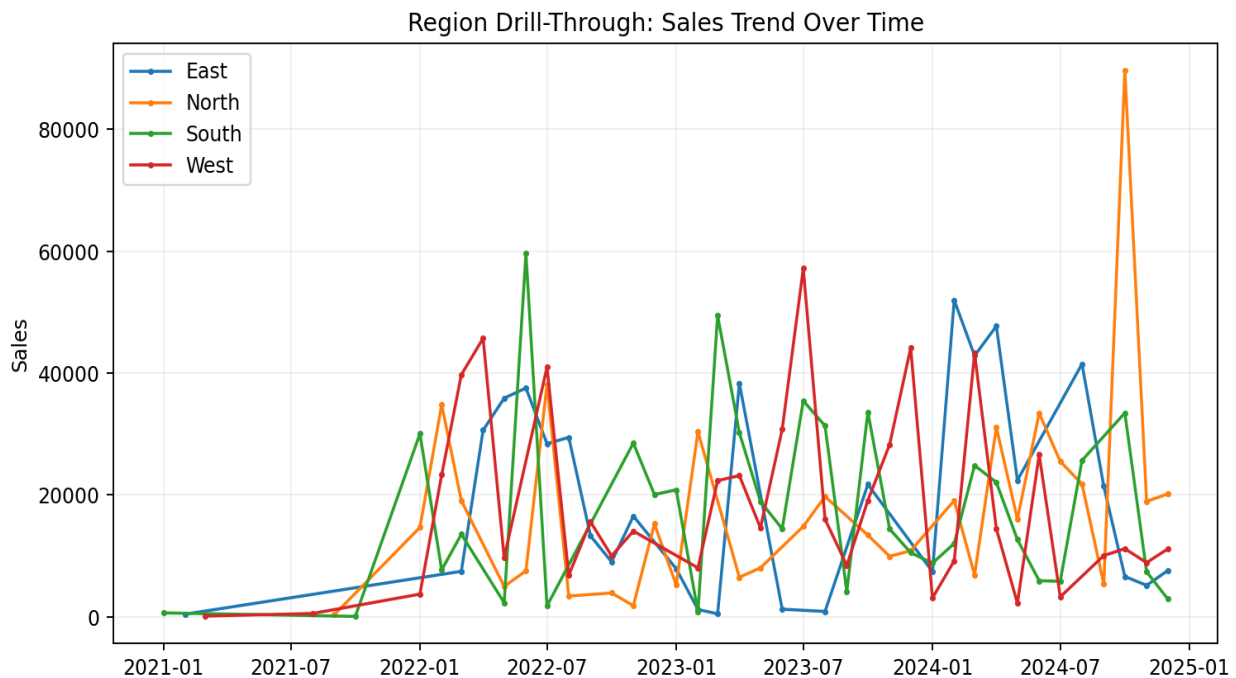
A Region Details page is created with Region in the Drill-through field well and a line chart of Date versus Sales. Selecting a Region from the main page and drilling through opens the trend for that region only.

Power BI implementation

1. Create a page named Region Details.
2. Add Region to Drill-through filters.
3. Add a Line Chart with Date Hierarchy on X-axis and Sales in Values.

4. Add a Back button for navigation.

Dataset result: Overall region totals: West Rs. 625,797, South Rs. 590,265, North Rs. 550,844, East Rs. 535,112.



Question 15: Design a fully interactive Power BI dashboard using Slicers, Filters, Drill-Down, and Drill-Through to explore sales data.

Solution

The final dashboard combines summary cards, Sales by Product, Sales Trend, Sales by Region, Category and Region slicers, report-level Country filtering, hierarchy drill-down, and Product/Region drill-through detail pages.

Power BI implementation

1. Add KPI cards for Total Sales and Order Count.
2. Add Region and Category slicers.
3. Add Pie/Bar charts for Product, Region and Category analysis.
4. Add a Date hierarchy line chart with Drill-Down.
5. Apply Country as a Report-Level Filter where required.
6. Create Product Details and Region Details Drill-Through pages.
7. Enable cross-filtering between visuals and add Back buttons on detail pages.

Dataset result: The complete dataset contains 250 orders and total sales of Rs. 2,302,018. This design provides interactive summary-to-detail analysis.

Final Power BI Configuration Summary

Feature	Implementation
Basic Filter	Date Year = 2023
Report-Level Filter	Country = India
Slicers	Region and Product Category
Drill-Down	Country -> Region -> Category; Year -> Quarter -> Month -> Day
Drill-Through	Product Details and Region Details pages
Measure Filter	[Total Sales] > Rs. 50,000
Measure	Total Sales = SUM('Sheet1'[Sales])